

Disclaimer

IMPORTANT SAFETY WARNING & DISCLAIMER

THIS PROJECT INVOLVES WORKING WITH HIGH VOLTAGE (230VAC) . HIGH VOLTAGE CAN CAUSE SEVERE INJURY, ELECTROCUTION, OR DEATH. FIRE HAZARD IS ALSO PRESENT.

This project is provided "**as is**", for educational and hobbyist purposes only. By using any part of this project (including hardware designs, PCB layouts, and software), you acknowledge and agree to the following:

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3. **Assembly & Modification:** Any errors in assembly, soldering (e.g., solder bridges), component selection, firmware modification or others actions are the sole responsibility of the user.
4. **No Warranty:** There is no guarantee regarding the accuracy of measurements, the stability of the software, or the safety of the hardware design for long-term use.
5. **Local Standards:** It is the user's responsibility to ensure that the device complies with local electrical codes and safety regulations.

IF YOU ARE NOT EXPERIENCED IN WORKING WITH MAINS VOLTAGE, DO NOT ATTEMPT TO BUILD THIS PROJECT.

Revision History

- 00: First release (30/05/2026)

Design Rationale & Project Overview

1.1 System Architecture

This project is a **Digital Power Meter** designed for real-time energy monitoring with acoustic alarm when a threshold is exceeded. The system architecture is based on the **ESP8266 (ESP-12F)** microcontroller, leveraging its Wi-Fi capabilities for IoT integration and an **ADS1115** to read voltage and current. For system health, an onboard **TFPT1206L1002DV temperature sensor** is included to monitor the internal thermal conditions of the enclosure. For local interaction, the device features a **0.96" color LCD display** that provides real-time data visualization and configuration menus. Users can navigate through multiple data pages and perform basic setup tasks using a **multi-function physical button**. Additionally, the system hosts a built-in **Web Server**, allowing for advanced remote configuration, firmware updates (OTA), and detailed system management via an intuitive **Web UI**.

The measurement stage utilizes two primary sensors:

- **Voltage Sensing:** Handled via a dedicated voltage transformer **ZMPT107-1 (TR1)** for galvanic isolation and safety.
- **Current Sensing:** Implemented using the **SCT-013** split-core current transformer (clamp meter).

Both sensors provide a 1VAC sinusoidal output. These signals are conditioned via a voltage divider network and sampled by an **ADS1115** 16-bit ADC, which communicates with the MCU via the **I2C protocol**.

1.2 Hardware Features & User Interface

To optimize GPIO usage, a **PCF8574 I/O expander** is employed for digital signal management and future use. The user interface consists of:

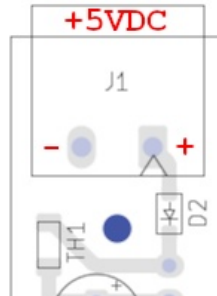
- **Buttons:** A multi-function navigation button (Short press: Page toggle | Long press: Confirm/Action) and a dedicated **Pin-hole Hardware Reset** button connected directly to Pin 1 (RST) of the ESP8266.
- **Acoustic Feedback:** An **active piezoelectric buzzer** triggers an audible alarm when a software-defined power threshold is exceeded.
- **Thermal Monitoring:** An onboard **TFPT1206L1002DV temperature sensor** (PTC thermistor) is integrated into the circuit. This component allows the system to monitor the internal temperature of the enclosure.
- **Display:** A **0.96" Color LCD** for local monitoring and basic interaction.
- **Web Server:** A Web User Interface for remote monitoring and full system configuration.

1.3 Power Supply & Protection

The device supports 2 power input options and 1 input 230VAC for Voltage reading:

1. **Terminal Block (J1):** Accepts a +5VDC regulated supply.

- **Warning - No reverse polarity protection:** This input currently lacks reverse polarity protection; correct orientation is mandatory.

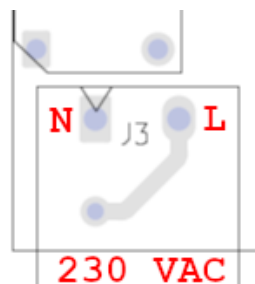


2. **USB Connector (J2):** For secondary power or debugging.

- **Communication & Debugging:** This interface utilizes a **CH340G** USB-to-Serial bridge and a **UMH3NFHATN** transistor pair to communicate with the **ESP8266**. This setup enables full debugging, serial communication, and firmware programming (automatic reset/bootloader entry).
- **Safety Feature:** The board implements a dual Schottky diode protection circuit on the positive rails of both J1 and J2. This prevents back-feeding, though it is recommended to disconnect J1 when using USB power.
- **Overcurrent Protection:** The +5VDC main rail is protected by a MF-NSMF050-2 PTC (Resettable Fuse) rated at 500mA.

3. **AC Voltage Input (J3):** High-voltage input for the ZMPT107-1 transformer.

- **Polarity Recommendation:** Although the transformer operates on AC, it is strictly recommended to respect the **Phase (L)** and **Neutral (N)** markings.



- **Safety Rationale:** The protection fuse is located exclusively on the Phase line. Inverting the polarity would mean that, in the event of a fault or a blown fuse, the Neutral line would be disconnected while the **entire internal circuit remains energized (live)** through the Phase line. **This creates a severe electrocution hazard even when the device appears to be non-functional.**
- **Safety Design:** The circuit features a **250mA T (Slow-blow) fuse** placed exclusively on the Phase line. Additionally, the line is

equipped with **three resistors in series** designed to limit the current to 1mA RMS at 230V, providing an essential layer of protection for the sensing stage.

1.4 Operational Limits & Grid Compatibility

This device is specifically designed and calibrated for **230VAC/50Hz** Phase-Neutral grids (Standard European Grid).

- **Voltage Specificity:** The internal components, including the protection resistors and the ZMPT107-1 transformer ratio, are optimized for 230V. Use with different mains voltages (e.g., 110-120V) may result in inaccurate readings or insufficient signal-to-noise ratio.
- **Frequency Specificity:** The system is tuned for **50Hz** operation. While the magnetic components (TR1 and CT) can physically handle 60Hz, the firmware's sampling window and RMS calculation algorithms are optimized for a 20ms period.
- **Warning:** Operation on different grid standards, high-voltage industrial lines, or Phase-to-Phase configurations (without a neutral line) may exceed the safety ratings of the components and the fuse, leading to hardware failure or fire hazard.

2.0 Connectivity & Software Integration

The firmware architecture features a built-in Web Server for real-time monitoring, local configuration, and connections management (Wi-Fi and MQTT). The device natively supports the MQTT protocol for a smart home integration with platforms such as Home Assistant.

2.0.1 LCD User Interface (Vertical Orientation)

The 0.96" display provides a multi-page interface for local data visualization and system control. Navigation is handled via the multi-function button (Short press: Page toggle | Long press: Confirm/Action).

- **Page 1: Real-time Monitor**
 - **Power:** Current power consumption displayed in **kW**.
 - **Temperature:** PCB temperature in **°C**.
 - **Graph:** A rolling history graph showing power trends (approx. 1-minute window).
 - **Status Badges:** Dynamic icons for **Wi-Fi** connection status (Green: Connected | Blue: Access Point Mode) and **MQTT** connection status (Green: Connected to server | Grey: Disconnected from server).
- **Page 2: Power Threshold Configuration**
 - Displays the active alarm threshold (Watts).
 - Allows **cyclic adjustment** of the new threshold (100W increments) via long-press. The value wraps around to zero once the upper limit is reached.
- **Page 3: Voltage Monitoring**
 - Displays the real-time mains voltage (**V**) as measured by the ZMPT107-1 transformer.
- **Page 4: Acoustic Alarm Settings**

- Displays the current buzzer mode: *Continuous*, *Slow Intermittent*, or *Fast Intermittent*.
- Users can cycle through these modes via long-press.
- **Page 5: Save Settings**
 - Summarizes the current threshold and buzzer configurations.
 - **Action:** A long-press commits these settings to non-volatile memory. *Note: Unsaved changes will be lost upon reboot.*
- **Page 6: System Information**
 - Displays network and build details: **Hostname**, **SSID**, **IP Address**, and **Firmware Version**.
- **Page 7: System Tools**
 - Allows selection of three critical functions: **AP Mode** (force access point mode), **Reboot** device, and **Reset** (Factory Reset) device.
 - **Confirmation:** To prevent accidental triggers, these actions require a long-press with a visual incremental timer-bar.

2.0.2 Web Interface & Configuration

The device features a responsive Web UI for remote monitoring and full system configuration. Users can connect by a browser directly to device's IP and access to the Web User Interface.

The IP and SSID are visible on LCD's page 6.

- **Home Page (Real-Time Dashboard)**
The landing page provides an immediate overview of the device status, electrical parameters and connections status:
 - **Connectivity Status:** Three dynamic badges (AP Mode, Wi-Fi, and MQTT). Status is indicated by color: **Grey** for inactive/disconnected, **Green/Blue** for active/connected.
 - **Live Telemetry:** Displays **Power (W)**, **Voltage (V)**, **Current (A)**.
 - **System Uptime:** Tracks operational time in a "Days hh:mm:ss" format.
 - **Data Refresh:** The page features an **asynchronous auto-refresh** every 10 seconds, with a manual "Refresh" button also available.
 - **Navigation:** Direct links to the **Configuration Page** and the current **Firmware Version** footer.
- **Configuration Page**
This restricted area allows for deep customization of the device behavior and network integration:
 - **General Settings:**
 - **Hostname:** Defines the device identity on the network.
 - **Manufacturer:** Metadata used for network discovery and Home Assistant integration.
 - **Language:** Localization support (currently **IT** or **EN**).
 - **Alarm Threshold (W):** Software-defined limit for the acoustic buzzer.
 - **Buzzer Mode:** Selection between *Continuous*, *Slow Intermittent*, or *Fast Intermittent*.

- **Wi-Fi Setup:**
 - **Wi-Fi Credentials:** SSID and Password configuration.
 - **IP Assignment:** Toggle between **DHCP** (automatic) or **Static IP**. If Static is selected, conditional fields appear for **IP Address, Subnet Mask, Gateway, and DNS**.
- **MQTT Setup (Optional):**
 - Connection parameters including **Broker IP, Port**, and optional **Authentication (Username/Password)** for secured environments.
- **Buttons:**
 - **Save & Reboot (Red):** Commits changes to non-volatile memory and restarts the device.
 - **Reboot (Red):** Performs a standard hardware restart.
 - **Firmware Update:** Redirects to the OTA (Over-The-Air) update interface.
 - **Home:** Returns to the main dashboard.

Manufacturing & Assembly Notes

3.0 Hardware Configuration

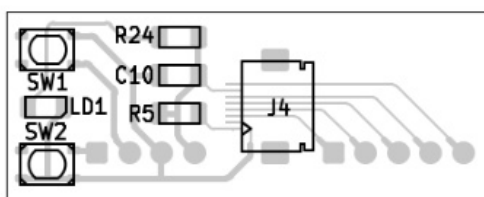
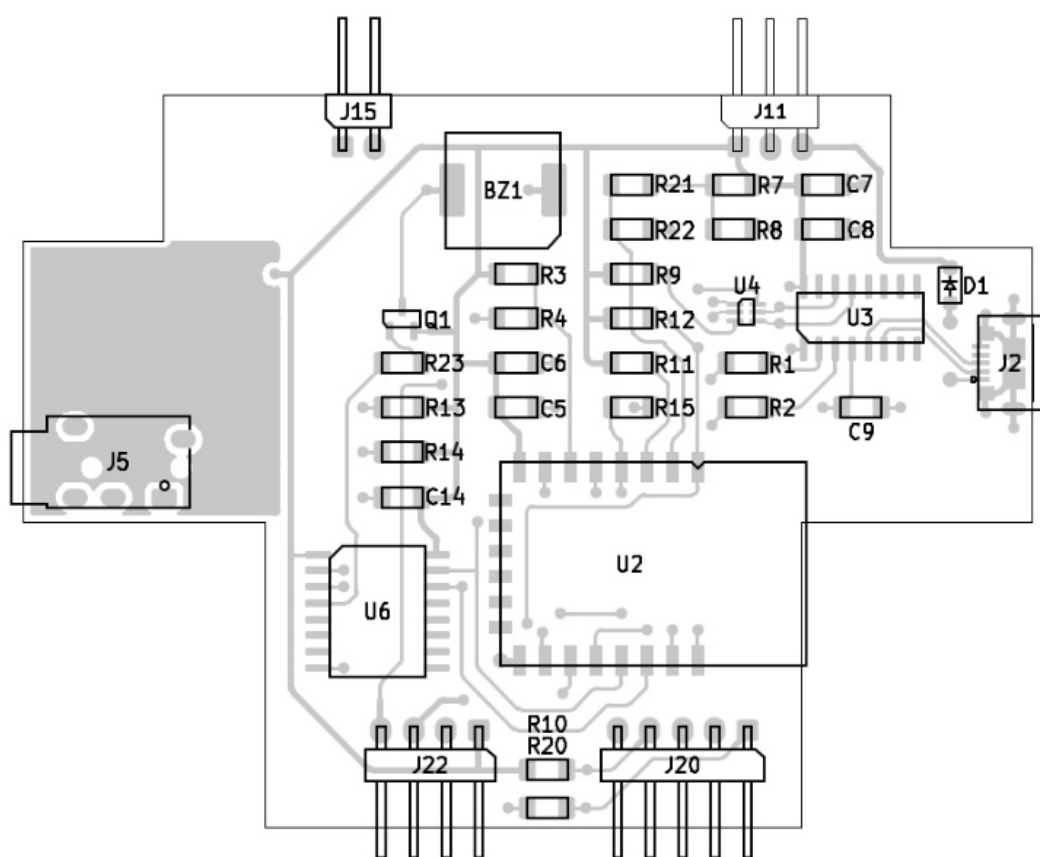
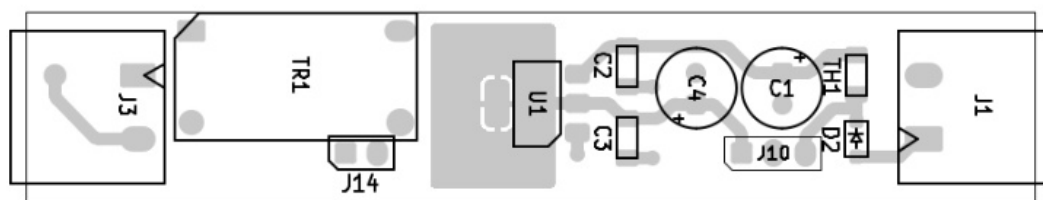
- **ADS1115 I2C Address Selection:** Is hardware-defined through two "0-Ohm bridge" resistors: **R16** and **R17**. These resistors act as jumpers to set the logic level of the **ADDR** pin (Pin 1) of the ADS1115. **Only one of these resistors must be populated at a time:**
 - **R16:** Connects the ADDR pin to **+3.3V**, ADS1115 Address **0x49**
 - **R17:** Connects the ADDR pin to **GND**, ADS1115 Address **0x48** (Default).

- **PCF8574 I2C Address Selection:** The I2C address of the PCF8574 is determined by the logic levels of pins A0, A1, and A2. With the default configuration where A0 is connected to +3.3V (High) and A1, A2 are connected to GND (Low), the resulting 7-bit I2C address is **0x21**.

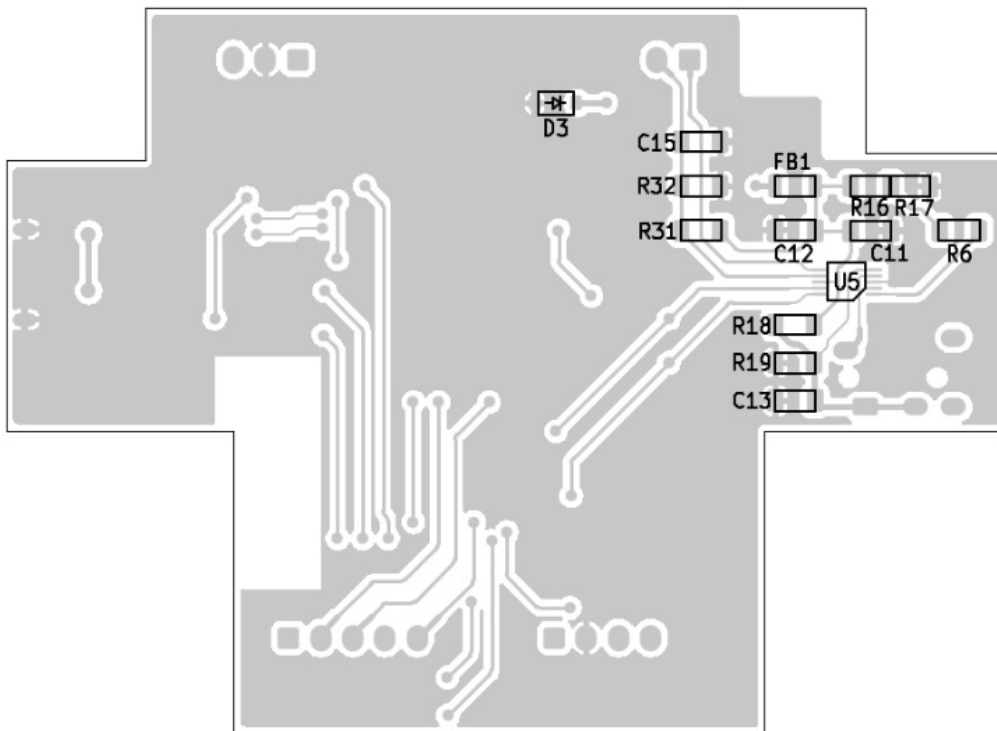
Note: The base address for the PCF8574 family is 0x20. The configuration [A2=L, A1=L, A0=H] corresponds to binary 0100 001, resulting in the hexadecimal address 0x21.

- **Safety Design:** To prevent assembly errors, the PCB layout utilizes a **shared common pad** for both R16 and R17. This physical constraint makes it nearly impossible to populate both resistors simultaneously, effectively protecting the +3.3V rail from a direct short circuit to Ground. However, be careful when assembling.

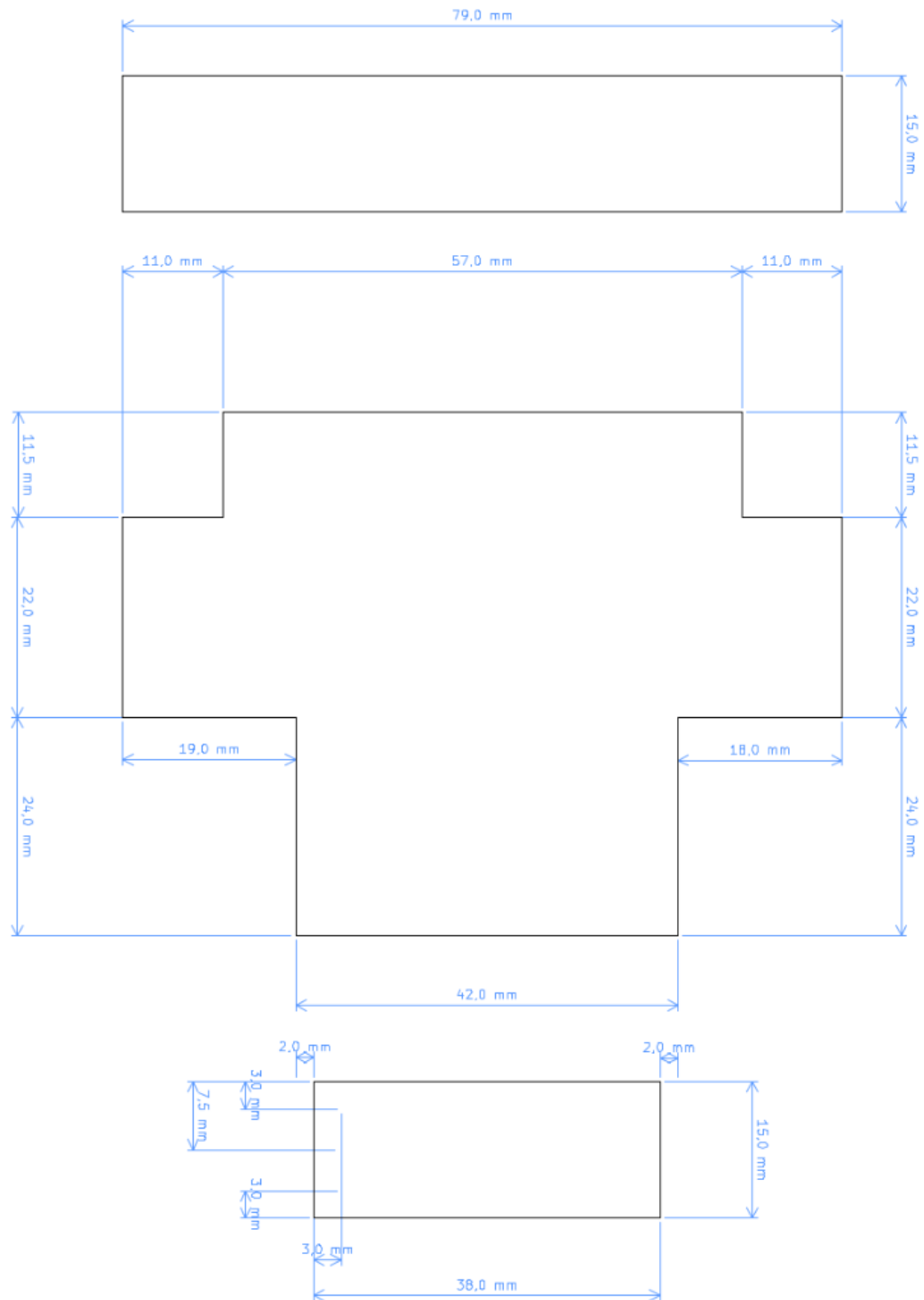
Assembly Layout: TOP Layer



Assembly Layout: BOTTOM Layer



Dimensions



4.0 Firmware

Peter The Power Meter has an embedded firmware designed for real-time single-phase energy monitoring. The system samples AC voltage and current waveforms, calculates Root Mean Square (RMS) values and active power consumption, provides a multi-page visual interface on a TFT display, and integrates with home automation systems via the MQTT protocol.

Technical Specifications

- **Language:** C++ (Standard C++11/14).
- **Framework:** Arduino Core for ESP8266.
- **Development Environment:** PlatformIO.
- **Core Hardware:** Espressif ESP8266 (e.g., Wemos D1 Mini).
- **Data Acquisition:** External ADS1115 16-bit ADC connected via I2C for high-precision differential sampling.
- **Sensor Support:** Current Transformers (SCT-013), Voltage Transformers (ZMPT107-1), and PTC temperature sensors (TFPT1206).

Key Features

- **High-Speed Sampling & RMS Calculation:** Performs rapid sampling of sinusoidal signals to calculate precise Voltage RMS, Current RMS, and Active Power (W) over a sliding 1-second window.
- **Display Management:** Features a multi-page User Interface on an ST7735 (0.96" TFT) display. It includes real-time power trend graphs, connectivity status badges, and system information pages.
- **IoT & Home Automation:**
 - **MQTT Telemetry:** Transmits energy data using a time-weighted average calculated over the publishing interval (default 20s). This optimization minimizes network congestion while maintaining high data accuracy.
 - **Home Assistant Integration:** Fully supports Auto-Discovery, automatically creating sensors and configuration entities (Select/Number) within Home Assistant.
- **Embedded Web Server:** A built-in web portal protected by HTTP Basic Authentication allows users to configure WiFi credentials, MQTT settings, alarm thresholds, and perform fine sensor calibration (offset and scale).
- **Alarm System:** Manages a buzzer via a PCF8574 I/O expander with programmable modes (Continuous, Slow Pulse, Fast Pulse) triggered when power exceeds a user-defined threshold.
- **Reliability & Maintenance:**
 - **Resilient Connectivity:** Implements an exponential backoff strategy for WiFi and MQTT reconnection.
 - **OTA Updates:** Supports Over-The-Air firmware updates through the web interface.
 - **Localization:** Built-in multi-language support (English and Italian) using PROGMEM to optimize RAM usage.

4.0.1 Sensor Calibration & Physics Guide (include/config.h)

The following constants define how the raw electrical signals from the sensors are translated into human-readable values (Amps, Volts, Watts).

- **Current Conversion (FACTOR_A_V)**

Description: This is the scaling factor for the Current Transformer (CT). It converts the voltage measured by the ADC into RMS Amperes.

Formula: $I_{RMS} = V_{ADC} * FACTOR_A_V$

Configuration: The FACTOR_A_V of 25.0f is set based on the characteristics of the SCT-013 current transformer. This specific configuration implies that the CT, along with its internal burden resistor, is designed to output 1mA for every 25A flowing through the primary circuit. This 1mA current is then converted into a voltage signal readable by the ADC. Therefore, a 1V RMS signal measured by the ADC corresponds to 25A RMS in the primary circuit.

When to change: This constant must be updated if:

1. You replace the SCT-013 with a model that has a different Current/Voltage ratio (e.g., 30A/1V or 100A/50mA).
2. You add or change an external burden resistor in parallel with the CT output.

- **Voltage Conversion (FACTOR_V_V)**

Description: This factor scales the low-voltage signal from the ZMPT107-1 transformer back to the actual grid voltage.

Formula: $V_{Grid} = V_{ADC} * FACTOR_V_V$

Configuration: The ZMPT107-1 is a 1:1 voltage transformer. In this setup, the primary side of the ZMPT107-1 is connected to an equivalent resistance of 246KΩ (achieved by three 82KΩ resistors in series). The secondary side uses a 1KΩ burden resistor to convert the current back into a voltage, which is then biased for the ADC input. The FACTOR_V_V of 246.0f is derived directly from the ratio of the primary resistance (246KΩ) to the burden resistance (1KΩ). This means that for every 1V RMS measured at the ADC input, it represents 246V RMS in the grid.

When to change: This constant must be updated if you modify the scaling "divider":

1. If you change the series resistors on the Primary side (e.g., to support higher voltages or reduce heat).
2. If you change the burden resistor on the Secondary side (e.g., using a 470Ω or 2kΩ resistor instead of 1kΩ).

Note: Changing the bias resistors (the 10kΩ resistors providing the 1.65V offset) does not affect this factor, as the ADS1115 measures the signal differentially across the burden resistor, effectively ignoring the common-mode DC offset.

- **ADC Precision (ADS_GAIN & ADS_MULTIPLIER)**

Description: These settings configure the ADS1115 (16-bit ADC) range and resolution.

Technical Detail: GAIN_TWO sets the full-scale range to ±2.048V. Since the ADC is 16-bit signed, the multiplier is calculated as $2.048 / 32768 = 0.0000625$.

Important: Do not change these. The hardware is optimized for this range. Increasing the gain will cause the signal to "clip" (saturate) at high power loads, leading to incorrect readings.

- **Sampling Rate (ADS_SPS)**

Description: Sets the sampling speed to 860 Samples Per Second.

Why it matters: To calculate an accurate RMS value for a 50Hz or 60Hz AC wave, we need to take as many "snapshots" of the wave as possible. 860 SPS is the maximum stable rate for the ADS1115 and provides high fidelity for power calculations.

- **Noise and Safety Thresholds**

CURRENT_NOISE_FLOOR (0.002f): This is a software filter. If the measured voltage from the CT is below 2mV, the firmware forces the current to 0.0A. This prevents the display from showing small "ghost" consumption caused by background electromagnetic noise.

VOLTAGE_MIN_THRESHOLD (50.0f): If the measured voltage is below 50V, the system assumes the voltage sensor is either faulty or disconnected. In this case, it falls back to the DEFAULT_GRID_VOLTAGE (230V) to allow power estimation based on current alone.

- **Temperature Monitoring**

PTC_AVG_SAMPLES (10): The ESP8266 internal ADC (used for the PTC temperature sensor) can be noisy. We average 10 readings to provide a smooth and stable temperature value.

TEMP_WARNING_THRESHOLD (45.0f): This is a safety limit used for reminder. If the internal temperature of the device exceeds 45°C, the display text turns RED to alert the user of potential overheating.

Summary Table for Quick Reference

Constant	Default Value	Unit	Description
FACTOR_A_V	25.0	A/V	Ratio of Amps per Volt from the CT circuit.
FACTOR_V_V	240.0	V/V	Ratio of Grid Volts per Volt from the VT circuit.
ADS_GAIN	GAIN_TWO	-	Sets ADC input range to $\pm 2.048V$.
ADS_SPS	860	Hz	Samples taken per second from the AC wave.
CURRENT_NOISE_FLOOR	0.002	V	Minimum voltage to consider a valid current reading.
VOLTAGE_MIN_THRESHOLD	50.0	V	Voltage below which the system uses a fallback value.

4.0.2 Per-device calibration limits (include/config.h)

These constants define the acceptable range and default values for the calibration parameters that are stored in EEPROM and applied at runtime. These parameters allow for fine-tuning sensor readings to compensate for component tolerances and environmental factors, ensuring maximum accuracy.

- **Default Multiplicative Scale (CAL_SCALE_DEFAULT)**

Description: This defines the default value for all multiplicative scale factors (voltage, current). A value of 1.0f signifies no correction, meaning the raw sensor reading (after initial hardware scaling) is used directly.

Usage: When the device is first initialized or after a factory reset, calibration scale factors will revert to this default.

- **Scale Factor Limits (CAL_SCALE_MIN, CAL_SCALE_MAX)**

Description: These constants set the minimum (0.1f) and maximum (10.0f) allowed values for any calibration scale factor (e.g., calScaleV, calScaleA).

Purpose: They prevent users from entering extreme or physically impossible scale values through the web interface, safeguarding the system from erroneous readings or instability.

- **Voltage Offset Limits (CAL_OFFSET_V_MIN, CAL_OFFSET_V_MAX)**

Description: These define the permissible range for the voltage offset calibration parameter (calOffsetV). The offset is applied after the scaling factor.

Range: From -500.0f to 500.0f Volts.

Purpose: This allows for small additive corrections to the measured voltage, compensating for minor discrepancies in the voltage sensing circuit that are not linear across the entire measurement range.

- **Current Offset Limits (CAL_OFFSET_A_MIN, CAL_OFFSET_A_MAX)**

Description: These define the permissible range for the current offset calibration parameter (calOffsetA). The offset is applied after the scaling factor.

Range: From -100.0f to 100.0f Amperes.

Purpose: Similar to voltage offset, this corrects for any baseline inaccuracies in current readings, particularly important at very low current levels to ensure a true zero reading when no current is flowing.

- **Temperature Offset Limits (CAL_OFFSET_TEMP_MIN, CAL_OFFSET_TEMP_MAX)**

Description: These define the permissible range for the temperature offset calibration parameter (calOffsetTemp).

Range: From -50.0f to 50.0f degrees Celsius.

Purpose: This allows for adjusting the reported temperature to match a known reference, accounting for variations in the PTC sensor or its placement.

Summary Table: Per-Device Calibration Limits

Constant	Default/ Range	Unit	Description
CAL_SCALE_DEFAULT	1.0f	-	Default multiplicative scale factor (no correction).
CAL_SCALE_MIN	0.1f	-	Minimum allowed value for scale factors.
CAL_SCALE_MAX	10.0f	-	Maximum allowed value for scale factors.
CAL_OFFSET_V_MIN	-500.0f	V	Minimum allowed voltage offset.
CAL_OFFSET_V_MAX	500.0f	V	Maximum allowed voltage offset.
CAL_OFFSET_A_MIN	-100.0f	A	Minimum allowed current offset.
CAL_OFFSET_A_MAX	100.0f	A	Maximum allowed current offset.
CAL_OFFSET_TEMP_MIN	-50.0f	°C	Minimum allowed temperature offset.
CAL_OFFSET_TEMP_MAX	50.0f	°C	Maximum allowed temperature offset.

4.0.3 PTC Calibration & Hardware (include/config.h)

The device monitors its internal temperature using a PTC (Positive Temperature Coefficient) sensor (specifically the TFPT1206 series). The sensor is part of a voltage divider circuit connected to the ESP8266's analog input (A0). The firmware performs a multi-stage calculation to reconstruct the actual resistance of the sensor and convert it into a precise temperature reading.

- **Voltage Source & Reference (PTC_V_SOURCE)**

Description: This is the regulated DC supply voltage used to power the temperature sensing divider.

Value: 3.3f (Standard logic voltage for the ESP8266).

When to change: Only if the sensing circuit is powered by a different voltage rail (e.g., 5V, though not recommended for this hardware).

- **Divider Pull-Up Resistor (PTC_R_PULLUP)**

Description: The fixed resistor R7 that forms the upper branch of the voltage divider with the PTC sensor.

Usage: It determines the voltage-to-resistance curve. A value of 33kΩ provides a good balance of sensitivity and range for the 10kΩ PTC sensor.

When to change: If you physically replace the R7 resistor on the PCB with a different value.

- **ADC Input Stage Compensation (PTC_R_DIVIDER_TOTAL & PTC_R_DIVIDER_LOAD)**

Description: These constants compensate for the voltage divider typically found on ESP8266 modules (like the Wemos D1 Mini) that maps a 0-3.3V input to the 0-1V range required by the internal ADC chip.

Technical Detail:

PTC_R_DIVIDER_TOTAL: The sum of the series resistor R21 and the load resistor R22.

PTC_R_DIVIDER_LOAD: The value of the load resistor R22 connected to GND.

When to change: Only if you are using a bare ESP-12F chip without an external divider on the ADC pin, or if your specific module uses different values (e.g., standard Wemos modules often use 220k/100k, resulting in a total of 320k).

- **Sensor Characteristics (PTC_NOMINAL_RES & PTC_COEFF)**

PTC_NOMINAL_RES (10,000):

The base resistance of the PTC sensor in Ohms at the reference temperature of 25°C.

PTC_COEFF (0.00411): The Temperature Coefficient of Resistance (TCR). It defines the linear slope of the resistance change per degree Celsius.

Formula: The temperature is derived using a linear approximation around 25°C: $\text{Temperature} = \{[(R_{\text{PTC}} / \text{PTC_NOMINAL_RES}) - 1] / \text{PTC_COEFF}\} + 25 + \text{Offset}$

When to change: If you use a different model of PTC sensor (e.g., a 1kΩ or 5kΩ version) or a sensor with a different TCR slope.

Summary Table: Temperature Sensing Constants

Constant	Default Value	Unit	Description
PTC_V_SOURCE	3.3	V	Logic source voltage for the sensing branch.
PTC_R_PULLUP	33,000	Ω	Fixed pull-up resistor (\$R_7\$) in the divider.
PTC_NOMINAL_RES	10,000	Ω	PTC resistance at 25°C.
PTC_COEFF	0.00411	-	TCR (Temperature Coefficient) of the sensor.
PTC_R_DIVIDER_TOTAL	100,220	Ω	ADC input stage total resistance (\$R_{21} + R_{22}\$).

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